

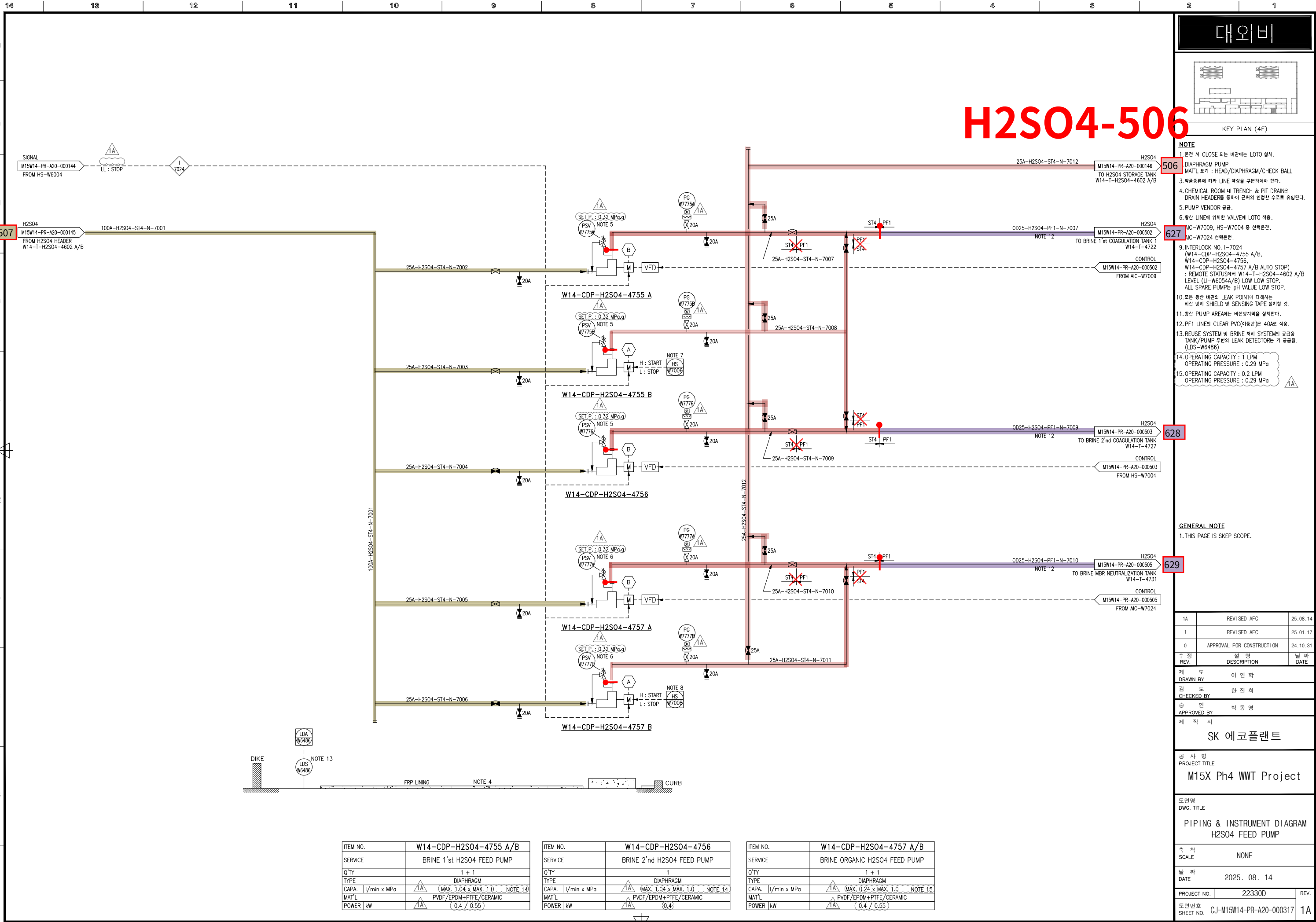
DRAWING LIST FOR TEST PACKAGE

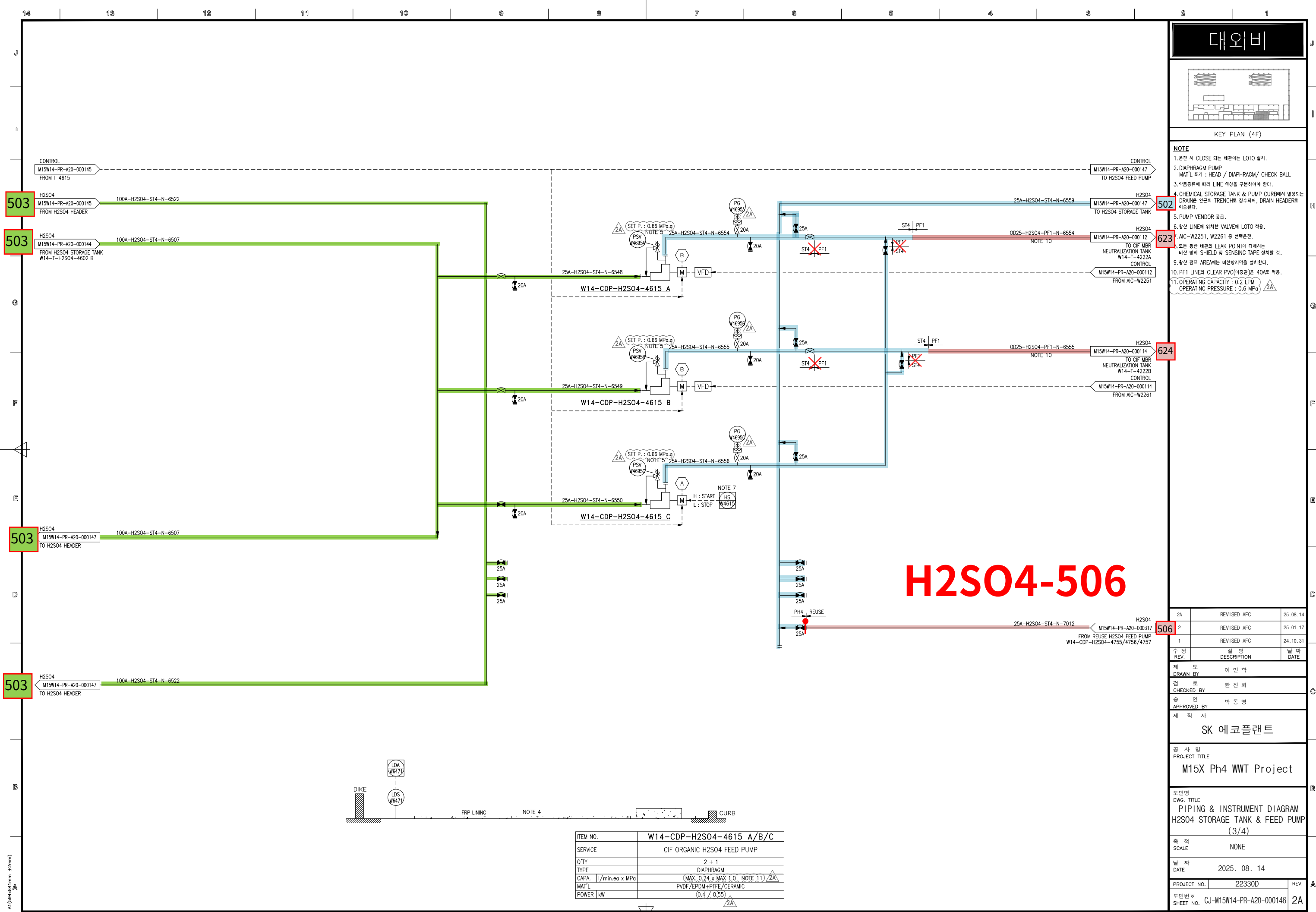
H2SO4-506

PROJECT NO.: 22328D

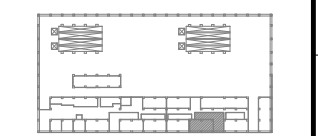
PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	506	4FA	25A-H2SO4-ST4-N-7007-01	1 OF 1	0	CJ-M15W14-PR-A20-000317/502		
2	506	4FA	25A-H2SO4-ST4-N-7008-01	1 OF 1	0	CJ-M15W14-PR-A20-000317		
3	506	4FA	25A-H2SO4-ST4-N-7009-01	1 OF 1	0	CJ-M15W14-PR-A20-000317/503		
4	506	4FA	25A-H2SO4-ST4-N-7010-01	1 OF 1	0	CJ-M15W14-PR-A20-000317/505		
5	506	4FA	25A-H2SO4-ST4-N-7011-01	1 OF 1	0	CJ-M15W14-PR-A20-000317		
6	506	4FA	25A-H2SO4-ST4-N-7012-01	1 OF 1	0	CJ-M15W14-PR-A20-000146/317		
7	506	4FA	OD25-H2SO4-PF1-N-7007-01	1 OF 2	0	CJ-M15W14-PR-A20-000317		
8	506	4FA	OD25-H2SO4-PF1-N-7009-01	1 OF 2	0	CJ-M15W14-PR-A20-000317		
9	506	4FA	OD25-H2SO4-PF1-N-7010-01	1 OF 2	0	CJ-M15W14-PR-A20-000317		





대외비



KEY PLAN (4F)

- NOTE**
1. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
 2. DIAPHRAGM PUMP MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
 3. 약품종류에 따라 LINE 색상을 구분하여야 한다.
 4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.
 5. PUMP VENDOR 공급.
 6. 황산 LINE에 위치한 VALVE에 LOTO 적용.
 7. AIC-W2251, W2261 중 선택운전.
 8. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
 9. 황산 펌프 AREA에는 비산방지막을 설치한다.
 10. PF1 LINE의 CLEAR PVC(이중관)은 40A로 적용.
 11. OPERATING CAPACITY : 0.2 LPM
OPERATING PRESSURE : 0.6 MPa

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 REV.	실명 DESCRIPTION	날짜 DATE

제 도 이 인 학
DRAWN BY
검 토 한 진 회
CHECKED BY
승 인 박 동 명
APPROVED BY

제 작 사
SK 에코플랜트

공 사 명
PROJECT TITLE
M15X Ph4 WWT Project

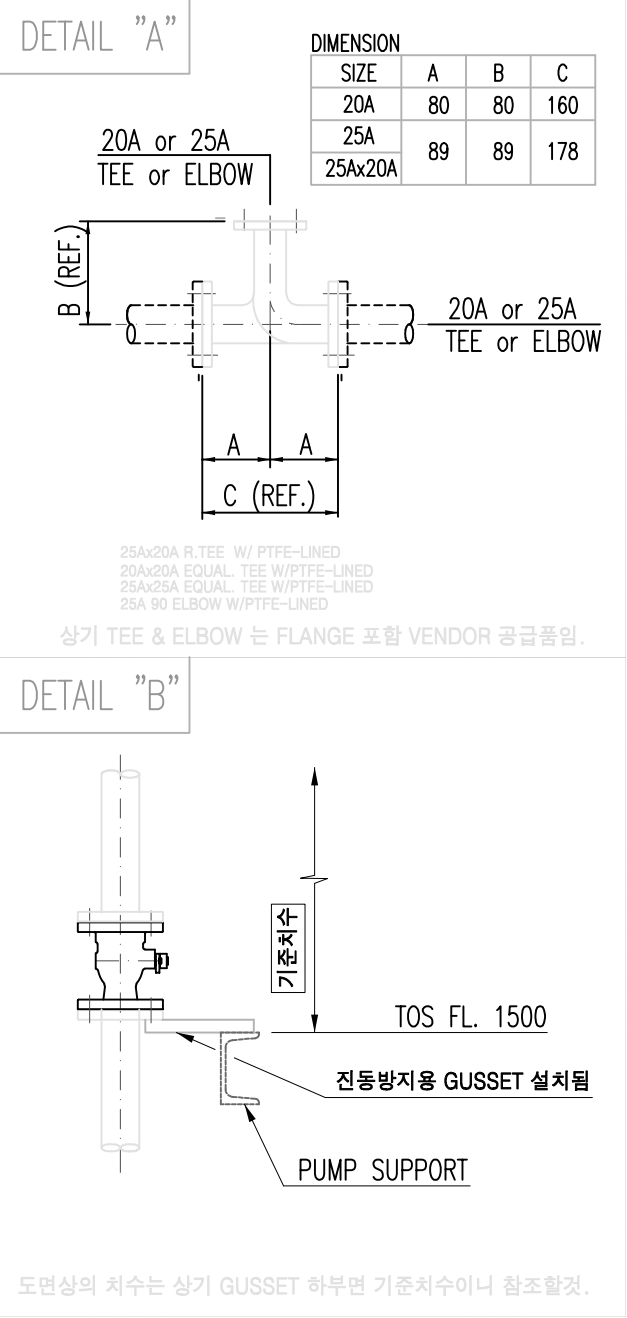
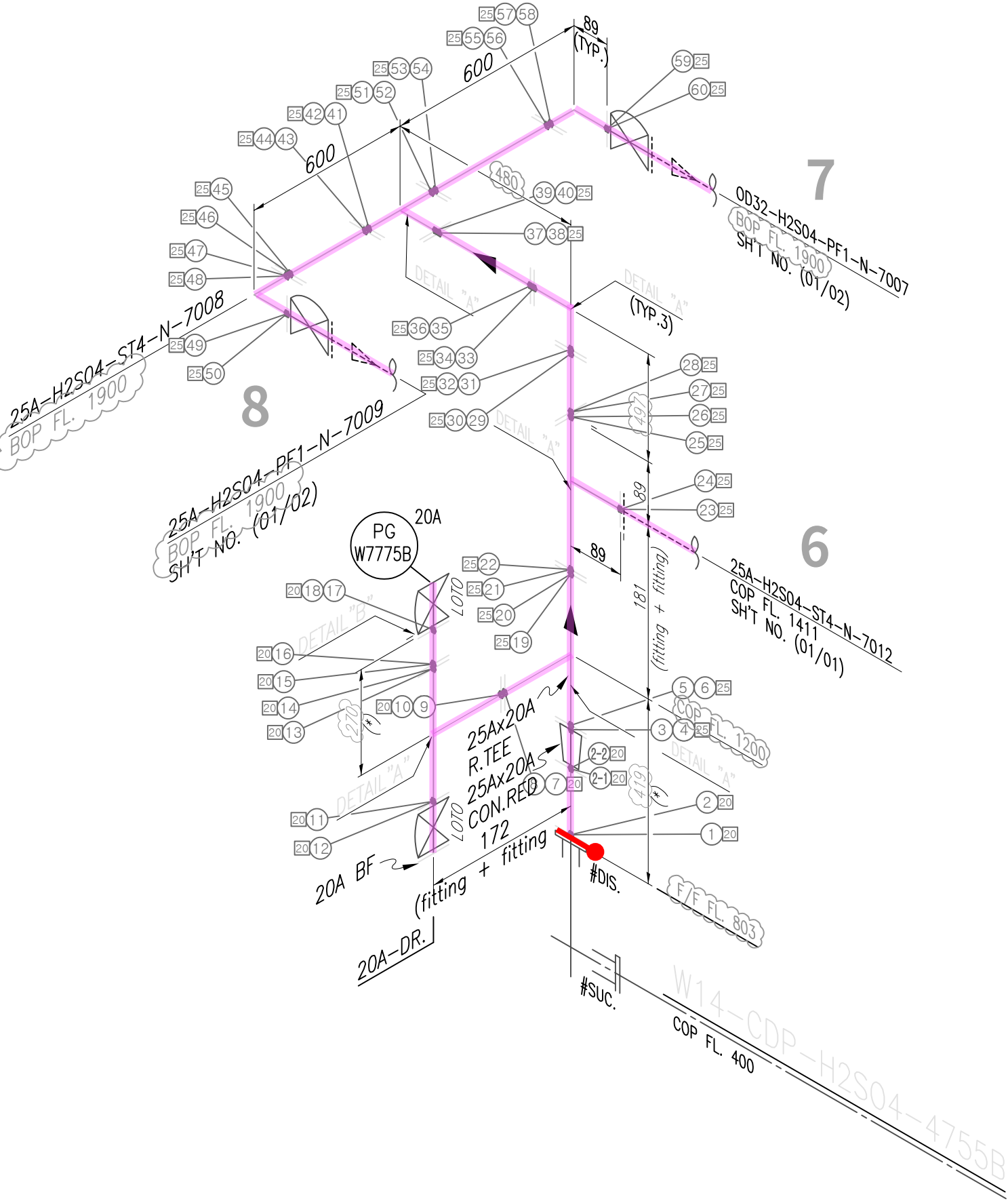
도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
H2SO4 STORAGE TANK & FEED PUMP
(3/4)

속 척
SCALE
NONE

날 짜
DATE
2025. 08. 14

PROJECT NO. 22330D REV. A
도면번호
SHEET NO. CJ-M15W14-PR-A20-000146 2A

ITEM NO.	W14-CDP-H2SO4-4615 A/B/C
SERVICE	CIF ORGANIC H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	l/min.ea x MPa (MAX. 0.24 x MAX 1.0. NOTE 11) 2A
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.4 / 0.55) 2A



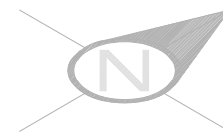
<NOTE>
1. (*) FIELD ADJUST

SK ecoplant			
ITEM	SIZE	Q'TY	DESCRIPTION
ST4			
P	25A	1.4M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S, With FLG
P	20A	0.2M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S, With FLG
E90	25A	3EA	ELBOW 90 DEG, STSF304+PTFE Lined, ERW – SCH 10S, With FLG
T	25A	2EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S, With FLG
T	20A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S, With FLG
TR	25Ax20A	1EA	TEE RED, STSF304+PTFE Lined, ERW – SCH 10S, With FLG
F	25A	23EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
F	20A	8EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
G	20A	6EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	13EA	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	20A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VDLO	25A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
VDLO	20A	2EA	KS 10K, RF, Teflon Lined SSC13, EPDM + PTFE Diaphragm, BB, HW
			+LOTO
SH	25A	13EA	SHIELD
SH	20A	5EA	SHIELD
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF
RE	25Ax20A	1EA	CON. RED, STSF304+PTFE Lined, ERW – SCH 10S

SK hynix				
CLIENT				
CONTRACTOR				
COMMODITY				
LINE NO.				
PROJECT NAME				
JOB NO.				
SH'T NO.				
REV.				

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

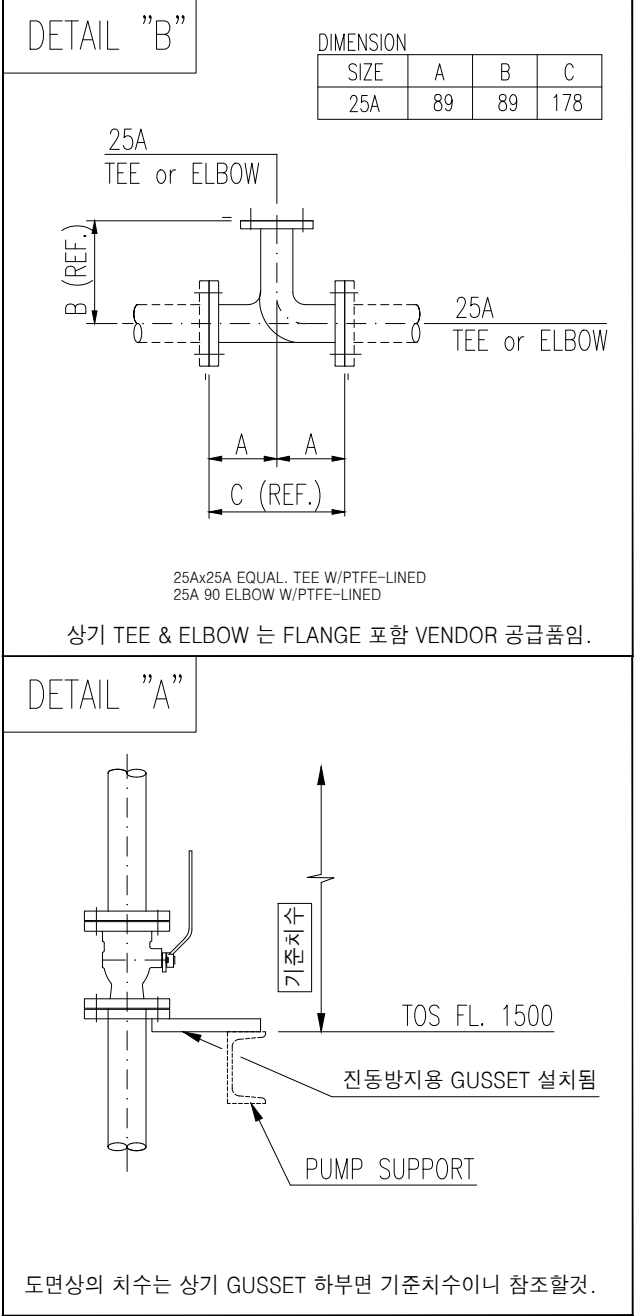
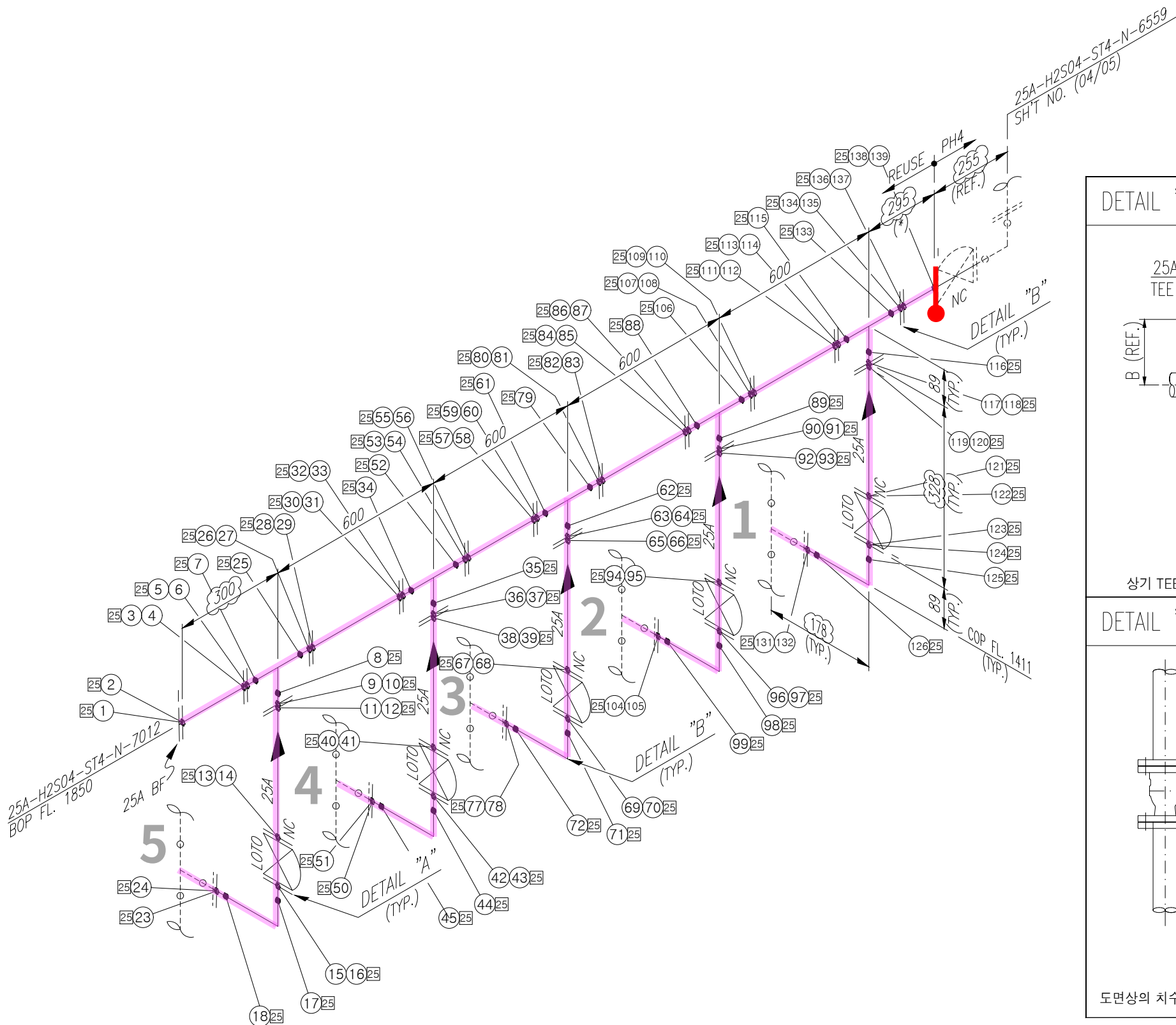
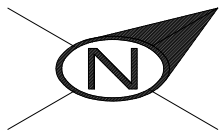
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	4FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-040013-A	REF. P. & ID	CJ-M15W14-PR-A20-000317	DATE	DESCRIPTION	BY	CHK	APP'D	PROJECT NAME	M15X Ph4 Project	JOB NO.	22330D	SH'T NO.	01 / 01	REV.	0
	THICK.	-	DESIGN	1.0	60									25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	H2S04							
			OPERATING	ATM	AMB														25A-H2S04-ST4-N-7008							



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

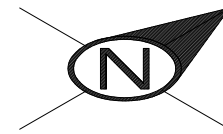
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	—	DESIGN	1.0	60	UNIT	4FA						
			OPERATING	ATM	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-040013-A							
						CJ-M15W14-PR-A20-000317							
									25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



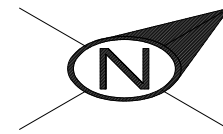
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR 	PROJECT NAME M15X Ph4 Project		
	THICK.	—	DESIGN	6.7	60	UNIT	4FA										
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A		△ REV. NO.	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H2S04	JOB NO. 22330D	
						REF. P & ID CJ-M15W14-PR-A20-000146								LINE NO.	25A-H2S04-ST4-N-7012		



CLIENT  7

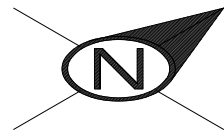
경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

CLIENT		8
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경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		PF1	
P	OD25	11.7M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	11.7M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	4EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
T1	OD25	1EA	TEE-EQUAL, PE, UNION, BACK RING(PTFE)
SC	40A	10EA	SOCKET, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
TB	40A	1EA	TEE-BOX
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
G1	OD25	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B1	OD25	1SET	HEX MACHINE BOLT & NYLON NUT :SM45C+HOT DIP GAL'V/W WASHERS

		ST4	
P	25A	0.6M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S, With FLG
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S, With FLG
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	5EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	5EA	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS

[illegible]

<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (*C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	-	DESIGN	3.3	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	3.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A	 REV. NO.	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY H2SO4	JOB NO. 22330D	
						REF. P & ID CJ-M15W14-PR-A20-000317		DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO. OD25-H2SO4-PF1-N-7010	SH'T NO. 01 / 02	REV. 0